

PEECHARA SATHWIK

Python Developer | AI & Data Analytics

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SUMMARY

Aspiring Python Developer with hands-on experience in NLP, Computer Vision, and AI-powered application development. Skilled in building end-to-end ML pipelines and integrating large language model APIs. Experienced in backend development, debugging, and developing scalable AI solutions.

EDUCATION

B.Tech in Computer Science (Data Science), Malla Reddy Engineering College, Hyderabad | 2023-2027
•**CGPA: 7.47 / 10**
Senior Secondary (XII), TSBIE - Science, Alphores Junior College Karimnagar | 2021-2023
•**Percentage: 70.4%**
Secondary (X), TSBSE, Alphores High School Karimnagar | 2021
•**CGPA: 10.00 / 10**

SKILLS

- **Languages:** Python, Java
- **Databases:** SQL: MySQL (RDBMS); familiar with NoSQL concepts
- **Cloud:** Google Cloud Fundamentals
- **AI / ML:** NLP, Transformers, Computer Vision, Machine Learning, Generative AI, Stable Diffusion, LoRA Fine-Tuning, Prompt Engineering
- **Libraries:** NumPy, Pandas, OpenCV, Pillow, Diffusers, Hugging Face Transformers
- **Tools:** Git, GitHub, Kaggle (cloud-based GPU training)

PROJECTS

- NexusAI – AI Chatbot with NLP & Multimodal Input** *Python | Flask | Groq API | Google Gemini API | MySQL | SQLAlchemy | NLP | PyMuPDF* 2026
- Full-stack AI chatbot: user input → Flask /chat route → Groq/Gemini (auto-failover) → real-time NLP analysis (intent/sentiment/language/entities) → MySQL logging (SQLAlchemy) → /analytics dashboard for usage trends.
 - Logs every conversation to MySQL with per-message NLP metadata and an /analytics endpoint for intent, provider performance, and usage trends; stress-tested with adversarial inputs (SQL injection, XSS, sarcasm, code-switching) to validate query safety and surface real NLP limitations.
- AI Manga Generation Pipeline – Text-to-Manga System** *Python | Flask | Stable Diffusion 1.5 | LoRA | Gemini API | OpenCV | Pillow | Hugging Face* 2025 – 2026
- Text-to-manga system: raw story text → Gemini (narrative beat extraction) → Gemini (20-rule panel layout engine) → LoRA-fine-tuned Stable Diffusion 1.5 (panel image synthesis) → OpenCV/Pillow (compositing into final page) → Flask web UI with live progress.
 - Fine-tuned a LoRA on the Manga109 dataset (15,500 steps, dual T4 GPUs) for manga-style synthesis, producing full 1800×2400px pages via a 3-stage pipeline with real-time SSE progress tracking in the browser.
- Codex Council – Multi-Agent LLM Code Review Pipeline** *Python | Ollama | Groq / Gemini / Mistral / OpenRouter APIs* 2026
- Multi-agent code-review tool: codebase → indexed (BM25 + vector embeddings) → Planner routes query to specialist teams (security, architecture, bugs, optimization) → each team runs Generator→Critic→Verifier debate → Mediator resolves cross-team conflicts → final report.
 - Runs on local Ollama models with automatic key rotation and multi-provider fallback (Groq → OpenRouter → Gemini → Mistral) for resilience, keeping the pipeline operational even when individual API keys or providers go down.

CERTIFICATIONS

- Azure AI Fundamentals (AI-900) — Microsoft
- Data Analytics Job Simulation — Deloitte (Forage)
- Machine Learning Certification — Tutedude (Nov 2025)
- Internship Certificate — AimR Edu (2024)
- Data Structures — TechAugusta (Oct 2024)
- Google Cloud Computing Foundations — Google (Apr 2026)